

General Relativity and the Probability Interpretation of Everett's Relative State Formulation

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(Dated: October 8, 2007)

One of the primary conceptual difficulties facing the multiple worlds interpretation (MWI) of quantum mechanics is the interpretation of the Born rule measure as a probability. Much less problematic would be to assign each of the N branches associated with a measurement the equivalent probability $1/N$, rather than the probability $|a_n|^2$ prescribed by the Born rule. In this paper, the aesthetic advantages of this alternative to the Born rule are discussed, and a modified version of the MWI is presented that incorporates this alternate probability rule, makes predictions consistent with quantum mechanics, and respects the full ontology of the original Everett programme. In addition, an outline for the derivation of this modified MWI from a background-independent field theory (such as general relativity) is proposed, making extensive use of the 4-geon particle model and the Feynman path integral technique, in a manner that does not require the Born rule as an independent postulate.

Keywords: multiple worlds interpretation, Born rule, probability interpretation, general relativity, geon.

I. INTRODUCTION

Of the various interpretational difficulties facing Everett's relative state formulation [1], a.k.a. the multiple worlds interpretation (MWI), perhaps the most critical is the interpretation of the Born rule measure as a probability. According to the Born rule, a "probability measure" $m_n = |a_n|^2$ is assigned to each of the N experimental outcomes (also termed "branches," "trajectories," or "worlds") associated with a measurement, with a_n being the complex coefficient associated with the n^{th} outcome, $n \in [1, N]$, as calculated by the Schrödinger equation. Wallace [2] has referred to this difficulty as the *quantitative problem*: why is the probability of a given branch quantified "according to the probability rule (i.e., the Born rule), and not (for instance) some other assignment of probabilities to branches?" Everett [1] and others (see, e.g., [3] [4]) have attempted to demonstrate that the Born rule arises naturally from the essential makeup of the MWI, and need not be assumed independently. Others, however, have criticised derivations of these types on the grounds that they contain hidden probabilistic assumptions that effectively "sneak" the Born rule into the formalism [5] [6]. Following this line of reasoning, it may be concluded that the application of the Born rule to the MWI constitutes an independent postulate that is separable from the remainder of the Everett interpretation.

Indeed, the inherent difficulty of deriving the Born rule from the MWI is underscored by the fact that the essential ontology of the MWI, according to which each world is equally "real," suggests an obvious alternative, according to which each branch is assigned the equivalent probability measure $m_n = 1/N$. This alternative rule, which will be referred to here as the "equal probability postulate" (EPP) (but has also been referred to as "outcome counting" [7], "egalitarianism" [8], or the "alternate projection postulate" [6]), preserves the unitary part of quantum theory [6], and reinforces the argu-

ment that the incorporation of the Born rule – or any alternative to the EPP, for that matter – into the MWI requires an independent assumption. Indeed, these assumptions may be hidden so cleverly that they take considerable effort to uncover. The derivation of the Born rule using Deutsch-Wallace decision theory, for instance, relies critically upon a postulate termed by Wallace [2] "measurement neutrality;" and measurement neutrality, in turn, necessarily *excludes* the EPP as a possibility, as discussed in Section 5.2 of [8]. As innocuous as measurement neutrality may appear at first glance, it comes at the cost, therefore, of violating the essential ontology of the equivalent "reality" of worlds.

The interpretational advantages of the EPP over the Born rule have long been recognized [9] [10] [11] [6]; as stated by Graham, "it is extremely difficult to see what significance such a measure [the Born rule measure] can have when its implications are completely contradicted by a simple count of the worlds involved, worlds that Everett's own work assures us must all be on the same footing" [9]. Indeed, the EPP may be the *only* solution to the quantitative problem: "The quantitative problem is perhaps unproblematic in the case of equally weighted superpositions, but has been considered fatal in the case of unequally weighted superpositions. ... [I]t seems that all Everett can say is that each outcome occurs in exactly one branch, which, we might think, will yield equal probabilities if any at all" [8]. In fact, the notion of "equally weighted superpositions" suggests that the motivation for the EPP is essentially an *argument by symmetry* – one that is made all the more powerful by virtue of the fact that it is independent of one's ontological stance regarding the "reality" of parallel worlds.

The difficulty with the EPP, of course, is that if one simply replaces the Born rule with the EPP, the resulting theory yields incorrect predictions. To resolve this issue, several independent EPP-based variations of the MWI have been put forth that suggest some *additional* modification so that quantum statistics may be restored to the overall programme. One such proposal is the "mangled worlds" model [12], according to which the memory of observers in "small" worlds is destroyed or somehow converted to remember events from "large" worlds. A second proposal [7] assumes nonlinearity

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in the time-dependent evolution of the observer state, as well as an exponential time dependence for the nonlinear effects.

Some would argue that the EPP and the Born rule are irreconcilable, and that the MWI should therefore be ruled out as an interpretation of quantum mechanics [13]. This argument simply ignores proposals such as the above, which serve to demonstrate – by example – the feasibility of recovering the Born rule on a coarse-grained level, via a programme that assumes the EPP at the fine-grained level. The difficulty of such programmes so far, however, is that it has proven difficult to find a deeper justification for their basic postulates (other than the EPP, which may be argued to represent a symmetry principle and need no further justification). Why do large worlds mangle small worlds? Why does the time dependence for nonlinear effects decay exponentially? The argument that “it works” is sufficient to give them provisional consideration; nevertheless, one would hope that some deeper principle may be found for their justification. It is not the argument of the present paper that this cannot be done; perhaps it can. Indeed, too little attention has been paid to efforts of this type.

The goal of this paper is to demonstrate, by example, that it is in fact possible to replace the Born rule with the EPP, to make correct predictions, to preserve the salient ontological features of Everett’s original proposal, and – most difficult of all – to point the direction for the derivation of the additional elements from deeper principles. In this proposal, those deeper principles are envisioned to be provided by an underlying field theory (such as general relativity) in a manner that preserves *its* most favorable ontological elements: e.g., background independence and locality. Indeed, many of the details of the proposed scheme have been guided by a considered respect for ontological concerns wherever they arise. The proposed scheme (modified MWI, or MMWI) will therefore be developed via a series of stepwise modifications to the standard MWI (SMWI), so that the ontological motivation behind each modification can (hopefully) be made clear.

The first step involves, simply, a replacement of the Born rule with the EPP, and Sec. II presents an argument for the aesthetic superiority of the EPP compared to the Born rule. Next, Sec. III presents how quantum statistics is restored via a reconceptualization of the number of branches associated with a given quantum mechanical measurement: rather than associating *one* branch with each experimental outcome, there are assumed to be *multiple* branches with each individual outcome, with the number of branches being proportional to $|a|^2$. Sec. IV presents an ontological argument that for these multiple branches to be considered distinct, they must represent *physically distinct observer states*. Therefore, a mathematical representation of these physical states will be sought, and it will be argued that general relativity satisfies the necessary requirements for such a representation. In particular, the concept of a “multiplicity” of solutions to general relativistic systems that admit closed timelike curves is reviewed, with emphasis on the natural applicability of the EPP to the probabilistic interpretation of “multiplicity.” The notion that “all is geometry” is also discussed, whereby all material objects (particles) are modelled as topologically nontrivial volumes of spacetime referred to in the literature as “4-geons” [14]

[15]. It will be shown in Sec. V that a small number of constraints over the mathematical representation of geons, and the spacetime in which they live, may be identified from which the requisite number of branches associated with a given measurement may be derived, and quantum statistics recovered. This demonstration makes heavy use of the Feynman path integral (FPI) formulation of quantum mechanics. In this manner, a general framework for the derivation of quantum statistics from general relativity – one that takes ontological guidance at many turns from Everett – is envisioned, one that does not require the Born rule as an independent postulate. Sec. VI presents ideas on how the mathematical foundation of the MMWI may be developed in greater detail, and Sec. VII contains a summary discussion and presents areas of future work.

II. MOTIVATION FOR THE EPP: THE PROBABILITY CRITERION

As stated above, the primary motivation for the EPP is that it provides the most “natural” probability rule that one may imagine upon first encountering the MWI. A simple example may be used to illustrate this argument. (The analysis that follows recapitulates an argument by Graham [9], and introduces some notation that will be continued through this paper.) Consider an experiment in which M identically prepared spin-1/2 particles are prepared, with their spins being measured sequentially. Using the Born rule, each particle is predicted to be observed to be spin up or down with probabilities m_{up} or m_{down} , respectively. Since each individual spin measurement produces two separate branches (worlds), there will be a total of 2^M worlds at the end of the M measurements. In each of these worlds, upon the completion of the M measurements, the observer is imagined to calculate the frequency p_{up} with which the particles were observed to be in the spin up state. In other words, m_{up} is what the theory predicts, and p_{up} is what is actually observed (with m_{up} being calculated once prior to experiment, and p_{up} being calculated separately in each of the 2^M worlds, after completion of the experiment). From a practical perspective, the prediction is tested by comparing the prediction m_{up} with the observation p_{up} (and likewise for spin down), using as large a value for M as is practically feasible. In particular, it is hoped that as $M \rightarrow \infty$, the physical measure of the number of worlds in which the Born rule appears to be false – that is, in which p_{up} deviates from m_{up} by an arbitrarily chosen (small) number δ – approaches zero.

The notion of testing a “probability rule” by comparing the predicted frequency to the observed frequency may be stated more generally for any arbitrary quantum mechanical experiment, using the following definitions:

Definition 1. *Experiments M .* M is the number of times that the experiment is run.

Definition 2. *Outcomes N .* N is the number of mutually exclusive possible outcomes for one experimental trial.

Note the assumption that the spectrum of experimental outcomes is *discrete*. This assumption may be made without loss

of generality, and is necessary for the purpose of defining a *measure* over the number of worlds in which a particular outcome is observed. A quantum mechanical experiment with a continuous spectrum of outcomes, such as a position measurement, should more properly be conceptualized as a theoretical (and perhaps unattainable) limit as the number of discrete elements of the position measurement apparatus approaches infinity.

Definition 3. *Frequency p_n .* For any individual world (of which there are N^M), $P_n \in [0, M]$ is the number times that the n^{th} outcome was observed, with $p_n = P_n/M \in [0, 1]$ being the frequency of this outcome among the M measurements.

Definition 4. *Probability measure m_n .* The “probability measure” of the n^{th} outcome, $n \in [1, \dots, N]$, is defined as m_n , which is the predicted probability frequency associated with the n^{th} outcome.

Note that p_n is an attribute of an individual world, whereas m_n is a predicted quantity that is independent of any individual experimental result.

Definition 5. *Error ϵ_n .* The difference between the predicted probability measure m_n and the observed quantity p_n will be referred to as the “error” ϵ_n , that is, $\epsilon_n = |p_n - m_n|$.

Definition 6. *Validity F_n .* The proportion $F_n \in [0, 1]$ of the N^M worlds in which the observer concludes that the probability measure is valid, as determined by the error being less than or equal to an arbitrarily chosen cutoff, $\epsilon_n \leq \delta$.

Given these definitions, the probability criterion may be stated in the following manner:

Axiom 1. *Probability criterion.* For any arbitrarily chosen δ , $\lim_{M \rightarrow \infty} F_n = 1$.

The probability criterion is essentially a mathematical statement of the notion that “most” of the worlds will produce an observer who concludes that the theoretical prediction is correct, to within some arbitrarily chosen cutoff. In a sense, the probability criterion may be interpreted as a *definition* of the very notion of probability: it is a requirement that must be met by the probability measure m_n . It therefore may come as a surprise that the probability criterion is not generally met if m_n is calculated by the Born rule, $m_n = m_n^{\text{Born}} = |a_n|^2$. That is to say that in *most* of the N^M worlds, the observer will conclude that the Born rule is false! The probability criterion is met, however, if m_n is calculated by the EPP, $m_n = m_n^{\text{EPP}} = 1/N$.

By way of illustration, suppose $m_{\text{up}} = 0.9$, $m_{\text{down}} = 0.1$, $\delta = 0.1$ and $M = 100$. These 100 measurements result in 2^{100} worlds, and it is asked: in what proportion of these worlds does the observer find that p_{up} falls within the interval $m_{\text{up}}^{\text{Born}} \pm \delta$? A quick calculation (using the formulae in the appendix to this paper) shows that the answer is a miniscule 5.58×10^{-8} percent. In contrast, the probability criterion is met if the probability measure is calculated using the EPP, $m_{\text{up}}^{\text{EPP}} = 1/N$. In this case, $m_{\text{up}}^{\text{EPP}} = 1/2$, and the proportion of worlds in which the observer finds that p_{up} falls within

the interval $m_{\text{up}}^{\text{EPP}} \pm \delta$ is a much larger 96.4 percent. It is simple to see that for any arbitrary δ , as M approaches ∞ , the former value approaches zero, and the latter approaches 100 percent.

It is not so difficult to see that the EPP is *generally* immune to the above difficulty. Indeed, it is easily demonstrated (see the appendix) that the EPP is a general solution to the probability criterion. The essential similarity between the EPP and the probability criterion is that in both cases, each of the N possible “branches” associated with a single measurement – or equivalently, each of the N^M distinct worlds resulting from M measurements – is considered to be, ontologically speaking, on an “equal footing.” The SMWI, on the other hand, does not seem to offer a clear ontological picture of the “reality” of alternate worlds. One is tempted to adopt a form of double-speak in the interpretation of the Born rule, whereby each world is “equally real,” with some worlds being “more real” (more probable) than others. The fact that the EPP, but not the Born rule, is a solution to the probability criterion is precisely what makes it the more natural choice for the probability measure.

III. THE ALTERNATE ENUMERATION POSTULATE AND THE BORN IDENTITY

Heuristically and in simple terms, the representation of an individual experiment by the standard MWI involves two steps. First, one must enumerate each distinct possible outcome of the experiment, i.e. the number N of “branches” associated with the measurement. Second, one must assign a probability measure m_n , with n an integer in $[1, N]$, to each distinct outcome using a probability rule such as the Born rule or the EPP. As discussed above, the EPP is presented as an aesthetically superior alternative to the Born rule within the second step. The straightforward way to incorporate the interpretational advantages of the EPP with the ability to make correct experimental predictions, therefore, is to modify the first step of the SMWI, referred to here as the “standard enumeration postulate” (SEP).

The SEP is deceptively simple. As described in Everett’s original paper [1], one begins (prior to measurement) with an observer in some definite physical state ψ^O , a system $\psi^S = \sum_n a_n \psi_n^S$ with eigenfunctions ψ_n^S , and a composite state ψ^{O+S} . The process of “observation” is defined as an interaction between the observer and the system that transforms the initial composite state ψ^{O+S} into the final state $\psi^{O+S'} = \sum_n a_n \psi_n^S \psi_n^O$. The notion of “branching” is introduced in the following manner: “[W]ith each succeeding observation (or interaction), the observer state ‘branches’ into a number of different states.” In other words, the concept of “branching” is defined such that the number of branches is equal to the number of final observer-states ψ_n^O .

However, Everett also makes the statement that “each branch represents a different outcome of the measurement ...” [1]. Here, Everett seems to imply that the number of branches is equal to the number of final states of the *observed system* ψ_n^S – as opposed to the number of states of the *ob-*

server ψ_n^O . Denoting the former by N_S and the latter by N_O , the tacit assumption that can be easily overlooked is that $N_O = N_S$. Furthermore, it is assumed that there is a one-to-one correspondence between the states ψ_n^O and the states ψ_n^S : “[T]he two have become correlated in a *one-one manner* ...” [1] (emphasis mine). This assumption of one-to-one correspondence will be referred to here as the “standard enumeration postulate” (SEP), and it is this assumption that will be replaced with an alternative way of counting observer states, which will be called an “alternate enumeration postulate” (AEP). In other words, the AEP is based upon the notion that N_O and N_S may be different. In keeping with the spirit of Everett’s original programme, the number of branches will be equated with N_O ; this seems only reasonable, since it is the observer who (by definition) calculates the frequency p_n and compares it to m_n , as described in the previous section. In general, *any* alternative to the SEP will be referred to as an AEP – although of course, the primary objective will be to identify a *particular* AEP that restores quantum statistics.

To illustrate the difference between the SEP and the AEP, consider once again the observation of the spin state of a single spin-1/2 particle with $m_{up}^{Born} = 0.9$. Prior to measurement, the system ψ^S can be expanded in terms of $N_S = 2$ eigenfunctions, $\psi^S = a_{up}\psi_{up}^S + a_{down}\psi_{down}^S$, with a_{up} equal to the complex square root of 0.9. According to the standard MWI, the process of measurement likewise transforms the observer ψ^O into $N_O = 2$ states, ψ_{up}^O and ψ_{down}^O . These two states are physically distinct by virtue of the fact that they are characterized by distinct internal physical recordings of the measurement result (for example, a bit in a computer’s memory). After the measurement, the observer will find himself in one of these two states with probabilities 0.9 and 0.1, respectively.

In the modified MWI, however, the EPP is adopted, according to which each state is equally likely. In addition, consider a simple AEP for determining the number of observer states after the measurement: assume that the process of measurement transforms the observer, not into $N_O = 2$ states, but rather into $N_O = 10$ physically distinct states: $\psi_{up,j_{up}}^O, \psi_{down,j_{down}}^O$, with j_{up} an integer in $[1, 2, \dots, 9]$ and $j_{down} = 10$. Since the number of distinct values of j_{up} is $J_{up} = 9$, and the number of distinct values of j_{down} is $J_{down} = 1$, then by the EPP, it is seen that the probability that the observer will measure spin up (or down) is 0.9 (or 0.1), in agreement with the prediction made by the standard MWI.

The world-splitting diagrams of the SMWI and the MMWI are illustrated in Figure 1 A and B, respectively. Figure 1 B illustrates the generalized assumption that J_n is proportional to $|a_n|^2$ – that is, it is proportional to the probability m_n^{Born} associated with the n^{th} eigenfunction of the observed system, as prescribed by the Born rule. The relation $J_n \sim |a_n|^2$ may be thought of as a specific implementation of the AEP and is referred to here as the “Born identity.” Just as the SMWI assumes quantum statistics by assuming the Born rule, the MMWI (in its simplest manifestation) assumes quantum statistics by assuming the Born identity. (A simple version of the many-minds interpretation [16] might simply stop right here and assert that the number of minds associated with any

measurement equals J_n , with J_n given by the Born identity. This interpretation, lacking any physical basis or deeper explanation, is inherently unsatisfying.) The goal of the remainder of this paper, therefore, will be to explore whether the Born identity may be derived from some deeper underlying framework, without simply assuming it outright.

The world-splitting diagram of Figure 1 B is the simplest one that one might conceive that gives rise to quantum statistics and fits the general scheme of the MMWI. However, other, more complex world-splitting diagrams could certainly be devised to meet these requirements. This is illustrated in Figure 1 C, in which the specific details of the world-splitting diagram are unspecified and are represented as a “black box.” As will be discussed in the next section, the precise configuration of the world-splitting diagram inside this “black box” is, in principle, a consequence of the mathematical representation of ψ^O , ψ^S , and ψ^{O+S} , as well as the rules that govern their evolution.

IV. GENERAL RELATIVITY AS A FOUNDATION FOR THE MMWI: CTCS, MULTIPLICITY, 4-GEONS, AND THE NOTION THAT “ALL IS GEOMETRY”

The reader may be wondering at this point whether there really is, after all, any substantive difference between the SMWI and the MMWI, or whether the difference is nothing more than notational reshuffling. A careful consideration of the essential structure of Everett’s original programme indicates that there is, in fact, a substantive distinction to be made. The issue at hand is whether there is any substantive difference between the claim that the measurement transforms the observer (once again using the spin-1/2 experiment as an example) into one of a total of $N_O = 2$ states, as in the SMWI, or – alternatively – $N_O = 10$ states, as in the MMWI. The argument that this is a *substantive* distinction between the SMWI and the MMWI is that each of these N_O final states is properly conceptualized as being *physically distinct*. In the words of Everett’s original paper [1], the final state is represented “in terms of a superposition, each element of which contains a *definite* observer state ...” (emphasis mine). Indeed, one could well argue that if the N_O states of the observer were not distinct *physically*, then it would make no sense to call them “distinct” at all. Therefore, it must be assumed that not only *up* and *down*, but also j_{up} and j_{down} , are *variables that represent some internal degree of freedom describing the physical state of the observer*. Thus, if the observer is a computer, then *up* and *down* represent bits in the computer’s memory that record the result of the measurement, while j_{up} and j_{down} represent some other – as yet undescribed – internal physical degree of freedom of the computer. The essential makeup of the Everett programme, therefore, requires the MMWI to demonstrate that the J_n states of the observer associated with the n^{th} outcome are, in fact, *physically distinct*. (One may ask whether the variable J_n (or the variable χ – see below) qualifies to be called a “hidden variable.” In one sense, yes: there is no way, using technology available at the time of this writing, to measure J_n . However, the notion that the J_n are physically distinct may be seen to

disqualify the application of the term “hidden” to the J_n .)

A physical interpretation of the variables N and J will require, of course, some sort of mathematical representation of the physical state of the observer in its initial state ψ^O , as well as a set of rules that predict its evolution from a single definite state ψ^O into one of a *multiplicity* N_O of distinct, alternative, mutually exclusive physical states ψ_n^O . It is interesting to point out that classical general relativity appears to give rise under the right conditions to situations in which a system in a single definite state has multiple distinct solutions to its time dependent evolution. An example of such a system that has been studied by several groups (see, for example, [17]) is that of a billiard ball whose trajectory is analyzed in a spacetime model that is not simply connected – in particular, one that admits closed timelike curves (CTCs). These authors discovered that it is possible for there to be *more than one* distinct solution to the billiard ball’s trajectory, each one of which is internally self-consistent. The number of distinct solutions to the billiard ball’s trajectory, i.e. the “multiplicity,” was of course dependent on the particular initial configuration of the composite system. Significantly, this multiplicity of solutions exists despite the initial conditions being specified in full.

In the realm of established classical physics, the existence of a multiplicity of solutions to the time-dependent evolution of a *fully specified* system appears to be unique to a model of GR that admits CTCs. (Statistical mechanics, for instance, does not provide a counterexample to this statement, since the initial state of a typical system in question is *not* fully specified.) By way of contrast, this sort of multiplicity is *not* characteristic, for example, of Newtonian mechanics. One might argue otherwise: for example, the Newtonian analysis of the trajectory of a pencil balanced on its tip might be expected to yield a multiplicity – in fact, an infinite number – of solutions to the direction of the pencil’s fall. On further reflection, however, it may be noted that Newtonian mechanics would predict that a *perfectly* balanced pencil would not fall at all, but would *remain balanced*. If one were to inject random (say, quantum) fluctuations to this model so that the pencil would fall, then the multiplicity of solutions to the pencil’s trajectory would be inherently attributable to quantum mechanics, not Newtonian mechanics.

It would be interesting, therefore, to attribute a probabilistic interpretation to the existence of this multiplicity of solutions via application of the EPP. In other words, ascribe to the observer ψ^O and system under observation ψ^S a general relativistic representation, one which admits CTCs, analogous to that given to the billiard ball and its environment discussed above. The “rules” that govern the evolution of these states are therefore nothing more than the laws of general relativity. The multiple distinct evolutions of the observer and system may be labelled $\psi_{n_o}^O, n_o \in [1, N_O]$ and $\psi_{n_s}^S, n_s \in [1, N_S]$, respectively. Via the probability criterion (the EPP), each of these alternative N_O evolutions of the observer is attributed an equal probability $1/N_O$.

The composite observer and system state ψ^{O+S} will likewise have a multiplicity N_{O+S} of solutions to its time-dependent evolution. Each of these solutions is internally consistent, but inconsistent with any other solution; thus, there

will be a separate 4-manifold W for the representation of each solution. Any pair of evolutions $\psi_{n_o}^O$ and $\psi_{n_s}^S$ are deemed “compatible” if they coexist within at least one W . Therefore, any individual of the N_S evolutions of the system must be compatible with at least one, but possibly more than one, of the N_O possible evolutions of the observer. This number is identified with the quantity $J_n, n \in [1, N_S]$ as discussed previously. (See Figure 1 B.) The quantity J_n (as well as N_O , which is the sum of the individual J_n s) is of course derivable, in principle, from the principles of general relativity, which govern this evolution and its associated multiplicities.

Interestingly, an independent programme for the “derivation” of quantum mechanics from general relativity has been proposed by Hadley [14] [15] that likewise relies in a fundamental sense on CTCs for its construction. In this programme, a particle at a given instant of time is modelled as a topologically nontrivial 3-dimensional volume a of spacetime (whose trajectory through time traces out a 4-dimensional volume of spacetime) termed a “4-geon,” with particular topological features being associated with particular particle types. In particular, the internal structure of a 4-geon is assumed to be non-causal, containing, for instance, CTCs as part of its essential makeup. In this model, physical objects are not envisioned as living “on top of” spacetime, but rather are *built out of* spacetime itself. This is the essence of the notion that “all is geometry” (and is also encapsulated in the phrase “background independence”). A specific 4-geon model of the electron, for example, has been proposed, according to which the non-causal structure of the 4-geon spacetime is seen to underlie the phenomena of spin-half [18] and charge [19]. The notion of modelling particles as regions of spacetime is not unique to Hadley. Similar attempts within the context of general relativity have been made, for example, by Einstein and Rosen [20], Wheeler [21], and more recently by Petrov [22]. Some alternative generalized field theories, such as that proposed by Evans [23], retain a similar notion of particles-as-geometry for their construction. Any direct connection between internal particle structure and quantum statistical laws, however – should such a relationship exist – is yet to be established.

In the next section, a general relativistic description of matter in terms of 4-geons will be assumed, and constraints over the manifolds W will be proposed, the result being a more complex MMWI model (Figure 1 E) that may be approximated by the more simple model of Figure 1 B and which is, therefore, consistent (at least in the approximation) with standard quantum theory. These constraints, termed here the “Born constraints,” will involve the definition of a parameter χ as an internal physical characteristic of a 4-geon, with $(\chi)^2$ playing a role analogous to that of J in the above discussion. (Compare Figure 1 B and 1 E). The overall goal is to demonstrate the feasibility of the following programme: to start with a generalized field theory, to derive from it a corresponding world-splitting diagram (i.e., one with the details of the “black box” of Figure 1 C filled in), and to demonstrate that its predictions match those of standard quantum mechanics. The Born constraints are presented, therefore, as an intermediate step in the construction of the world-splitting diagram from

an underlying field theory.

V. THE BORN CONSTRAINTS

The rules that govern the model of Figure 1 E will be referred to as the “Born constraints.” This model assumes a general relativistic description in terms of 4-geons of the initial states of the observer ψ^O (say, a computer), the system under observation ψ^S (the particle), and the composite observer-system state ψ^{O+S} . As discussed above, there will be multiple 4-manifolds W to represent the multiplicity of evolutions of the composite observer-system state ψ^{O+S} ; the ensemble of all such 4-manifolds, corresponding to a given experimental setup, will be labelled E .

In addition, assumptions will be presented concerning the actions of paths λ connecting two 3-regions a and b within a given W . These are to be interpreted as generic mathematical constraints to which each W is restricted. Although a and b are to be considered generic (but typically small) regions of spacetime, the special case may be considered that one or both correspond to a 4-geon. In this case, these assumptions form the basis for the construction of the physical descriptor χ , whose square will be shown to play a role analogous to that of J in the preceding discussion. In this manner, the Born constraints serve as a *generalized field-theoretic set of mathematical constraints*, from which the model of the MMWI as depicted in Figure 1 E may be built.

Consider a very general quantum mechanical experiment: given a particle that is observed within a 3-dimensional spacetime region s (corresponding to the location of the 4-geon upon emission by the source), calculate the relative probability that it will be detected within the region d_n (where d indicates particle detector, with n representing the n^{th} detector element). This experiment can be used, for example, to represent the electron 2-slit experiment, which Feynman has argued contains the “essential mystery” of QM [24]. The FPI technique for the analysis of this experiment will play a central role in the presentation that follows, and a compact review of this technique is presented for reference in Appendix B.

Following the general reasoning of Everett’s original programme [1], the observer ψ^O is defined to be in a single, well-defined initial state, and to have already observed and made *internal recordings* of the full experimental setup: the presence of the particle at the source s , a series of detectors at regions d_n , a system of barriers between source and detector, and the system under observation ψ^S in its initial state. It should be emphasized that these internal recordings are *assumed* in Everett’s original programme to represent an *accurate* representation of external reality. Indeed, an essential part of Everett’s programme is that the internal state of the observer ψ^O *determines* the state of the external world, including the experimental setup, to the degree that the external world has been observed and recorded. As stated originally by Everett, “the state of one subsystem [e.g., the experimental apparatus] does not have an independent existence, but is *fixed* only by the state of the remaining subsystem [e.g., the observer]” [1] (emphasis mine). Therefore, ψ^O may be used

as the boundary condition that *defines* the ensemble E of 4-manifolds $W \in E$. In other words, since any $W \in E$ contains (by definition) ψ^O , and it is assumed that ψ^O contains internal recordings of the experimental setup (including ψ^S), and it is further assumed by Everett that these internal recordings are *correct*, then any $W \in E$ *must also contain this full setup*. Thus, not only the initial state ψ^{O+S} , but also every possible solution to the evolution of ψ^{O+S} , is represented by at least one $W \in E$. (In contrast to the observer, the system under observation is not required to be in a *single* well defined state; rather, there may in fact be multiple initial states ψ^S , with different ψ^S s being represented by distinct W ’s. For example, the particle ψ^S , in its premeasurement state, may be in a well-defined location, but its physical orientation may differ between different W ’s.)

It may be remarked that this notion is in fact not so dissimilar in spirit to an emerging principle within quantum cosmologies known as the holographic principle, proposed originally by ‘t Hooft [25] and developed by Smolin [26], Susskind [27], and others, according to which the physical state of a system is encoded within the geometric structure of a boundary of that system. Although the usual statement of the holographic principle makes use of a 2-dimensional boundary, whereas ψ^O is not 2-dimensional, the spirit of the argument is the same: in the present construction, the observer ψ^O plays the role of such a boundary, and may be imagined as a holographic image of the outside universe.

Given a 3-dimensional volume a corresponding to a 4-geon at some instant of time, the variable χ will be defined in terms of paths connecting the 4-geon a to other regions of spacetime. Within any $W \in E$, given any two 3-regions a and b (where s and d are specific instances of the more general variables a and b), consider the set of equivalence classes of all possible paths connecting a and b (under the equivalence relation of homotopy), where $\lambda_{a,b}^i$ denotes a class representative of the i^{th} equivalence class. (Given the assumption that spacetime is characterized by multiple-connectedness at the small scale via the 4-geon particle model, there will be a very large number of paths that cannot be deformed smoothly into one another, and hence a very large number of distinct equivalence classes.) There is further assumed a suitable definition for the general relativistic action S^i of path λ^i . Assume that one method of approximating this set of paths in the non-relativistic limit (using a fixed, simply connected, flat background spacetime manifold M) is to enumerate the set of “all possible” paths $\varphi_{a',b'}^i$ from a' to b' in M (which map to a and b in W) by the technique of the Feynman path integral (FPI) [24]. (The mapping from W to M is discussed in further detail in Sec. VI B.) Furthermore, assume that the action of $\lambda_{a,b}^i$ in W is well-approximated in the non-relativistic limit by the action of the corresponding path $\varphi_{a',b'}^i$ in M as calculated by the FPI. (The equivalence of the action of $\lambda_{a,b}^i$ and the action of $\varphi_{a',b'}^i$ is also discussed further in Sec. VI B.) Since the distribution of the amplitudes of the Feynman paths φ on the unit circle in the complex plane (see Figure 1 D) are completely determined by the experimental setup (according to the standard technique of the FPI [24]), the same must also be true of the amplitudes of the paths λ , given the postulated relation-

ship between the φ and the λ .

Next, assume that given any set of paths $\lambda_{a,b}^i$ between a and b , there is a natural method of placing them into disjoint subsets whose union equals the entire set of paths, and with each subset containing k paths, $k \in [1, 2, 3, \dots]$. Thus, any individual path belongs to one and only one subset containing k paths, with k being called the “index” of the path; that is, the index is the number of paths with which any given path is coassociated (so that a path with index $k = 1$ is coassociated only with itself). The coassociation of these paths within subsets is assumed to depend upon the precise configuration of the region a . (See Sec. VIC for one possible method by which such coassociations may be generated.) If a and b represent two different locations in spacetime of the same 4-geon, it is assumed that the placement of paths $\lambda_{a,b}^i$ into subsets is preserved between a and b . Define $\chi_{a,b}^k$ as the total number of paths from a to b with index k . Replace a and b with s and d_n , respectively, and define χ_s^k as the sum of χ_{s,d_n}^k over all of the detector elements. Since s is interpreted as a 4-geon (corresponding to the state of the particle as it is emitted from the source), then χ_s^k is a *local parameter* describing the physical makeup of this 4-geon.

Given any individual subset of k paths, assume that there is a natural cyclic ordering from 1 to k such that two adjacent paths λ^i and λ^{i+1} are $2\pi/k$ out of phase, that is, $S^i - S^{i+1} = 2\pi\hbar/k + 2\pi\hbar m$ with m any integer. (This expression is derived from deeper principles in Sec. VIC. Also see Figure 1, D, which illustrates the case for a subset with index $k = 5$. Recall that the amplitude for the i^{th} path whose action is S^i is given by the expression $\phi^i = e^{-iS^i/\hbar}$.)

According to the FPI, the *total* amplitude ψ_{s,d_n} for a particle emitted from the source s to be observed in the region d_n is calculated as the sum of the individual amplitudes ϕ_{s,d_n}^i over all paths from s to d_n . Given the assumptions above, performing the summation using the MMWI paths λ (in place of the Feynman paths φ) will yield the same result ψ_{s,d_n} , since each path λ has the same action and hence the same amplitude as the corresponding path φ . Because of the cyclic ordering discussed above, it is plain to see that for $k > 1$, the amplitudes of the k paths within any individual subset will “cancel each other out,” i.e. they will sum to zero. (See Figure 1, D.) Thus, the paths with index $k > 1$ may be eliminated from the calculation of ψ_{s,d_n} , so that ψ_{s,d_n} is simply the sum over all paths with index $k = 1$.

Consider next the details of the “black box” in Figure 1, C. This will be assumed to take the two-step structure illustrated in Figure 1, E. (A heuristic justification for some of this structure is presented in Sec. VID.) By experimental design, the observer must record, at the beginning of the experiment, the emission of the particle from the source. This process of observation may feasibly be imagined to convey to the observer details regarding the internal structure of the 4-geon, among which is included the physical parameter χ_s^1 (the value of which may of course differ from one W to the next). (Note that since χ_s^1 is a local parameter, this does *not* require any non-local transfer of information!) It is assumed that this process of observation occurs in a manner that results in the world-splitting diagram as outlined in the first stage of

Figure 1 E, in which $\bar{\chi}_s^1$ indicates the minimum possible value of χ_s^1 . Since different W 's will contain a 4-geon with distinct values of χ_s^1 , there will be different branches to represent these distinct values (or ranges of values) for χ_s^1 . According to the EPP, each time the diagram “splits” into two directions, each of the two trajectories is attributed an equal probability measure. Therefore, the effect of this diagram is that lower values of χ_s^1 are *more probable* than higher values. (Note the assumption of transitivity to the issue of calculating probabilities involving successive branching events.) For the next stage of the world-splitting diagram, therefore, it may be assumed that χ_s^1 is approximately equal to its minimum value $\bar{\chi}_s^1$; branches representing higher, less probable values of χ_s^1 will be ignored for the sake of simplification (but are represented in Figure 1 E by dots “...”).

The minimization of χ_s^1 implies the minimization of χ_{s,d_n}^1 for each d_n . For χ_{s,d_n}^1 to take its lowest possible value, it is necessary for as many paths λ_{s,d_n}^i as possible to be placed into subsets of index $k > 1$, thus leaving the fewest possible number of paths to take index $k = 1$. Two previously stated requirements regarding the paths λ may be seen to confer a boundary on this minimum value. The first requirement is the one-to-one correspondence between the MMWI paths λ_{s,d_n}^i and the FPI paths φ_{s,d_n}^i ; the second is the requirement that the k paths within any individual subset must be cyclically ordered (Figure 1 D). First, it must be emphasized that the overall distribution of the amplitudes of *all* paths from s to d_n on the unit circle is determined by the experimental setup and may be calculated through application of the FPI. This overall distribution is *identical* for every $W \in E$. What is *different* from one W to the next is the way that these paths are arranged into subsets. This arrangement of paths into subsets cannot be done arbitrarily, however, given the above two requirements.

This may be communicated most effectively with a series of examples. Suppose hypothetically that half of the paths have phase $\pi\hbar$, and the other half have phase 0. In this case, it is possible to coassociate each individual phase 0 path with one phase $\pi\hbar$ path, so that all paths are of index $k = 2$, resulting in zero $k = 1$ paths and, therefore, $\bar{\chi}_{s,d_n}^1 = 0$. (See Figure 2, A.) On the other hand, suppose hypothetically that all of the paths from s to d_n have exactly the same phase, 0. In this case, none of them can be placed into subsets of $k > 1$, so that each path must be coassociated only with itself and therefore have index $k = 1$. Therefore, the value of $\bar{\chi}_{s,d_n}^1$ will be equal to the total number of paths from s to d_n . (See Figure 2, B.) As a third example, suppose a distribution such that the amplitudes of the vast majority (x_1) of paths are equidistributed throughout the unit circle, with a small remainder (x_2) of paths situated with phase 0. In this case, the vast majority may be coassociated into one very large subset with index $k = x_1$, leaving the remainder x_2 paths to take index $k = 1$, so that $\bar{\chi}_{s,d_n}^1 = x_2$. (See Figure 2, C.) This third example may be modified into a fourth one by considering the more complicated situation in which the distribution of the amplitudes of the x_1 paths is not uniform, but is nevertheless characterized by the property that its paths can be decomposed into subsets with k always greater than unity. This fourth example, like the third, yields $\bar{\chi}_{s,d_n}^1 = x_2$. (See Figure 2, D.)

It should be readily apparent that if a given manifold W fits into the category of this fourth example, then χ_{s,d_n}^1 takes its minimum value, $\bar{\chi}_{s,d_n}^1$: since all x_2 of the index $k = 1$ paths have the same phase, then there are no remaining paths with which they might be coassociated. Furthermore, it may also be noted that for a manifold that fits the category of the fourth example, the absolute value of the *sum of the amplitudes* of all index $k = 1$ paths – which was shown earlier to be equal to $|\psi_{s,d_n}|$ – is also proportional to the *number* $\bar{\chi}_{s,d_n}^1$ of index $k = 1$ paths being summed over, since they all have the same phase. Therefore, $\bar{\chi}_{s,d_n}^1 \sim |\psi_{s,d_n}|$, as illustrated in Figure 2 E and F. Squaring this expression yields $(\bar{\chi}_{s,d_n}^1)^2 \sim |\psi_{s,d_n}|^2$.

It may be argued that since the FPI takes into account *all possible* paths from s to d_n , then this “number of paths” is in fact infinite, rendering $\bar{\chi}_{s,d_n}^1$ infinite as well. However, a finite number of paths may be generated within the context of the FPI by considering a finite “representative” sampling of all possible paths, and it is argued that a similar technique of approximation (such as the lattice approximation technique discussed in Sec. VIB) may be applied to the enumeration of the MMWI paths λ_{s,d_n}^i . Nevertheless, it is proposed that an “exact” implementation of the MMWI will require an infinite number of paths, as is the case for the FPI, and that the MMWI technique is similar to the FPI technique in the sense that its predictions remain stable as the number of paths included in the calculations becomes infinite.

Although a full argument will not be presented here, it may be argued that the assumption of an infinite number of paths implies that it is always possible to categorize paths into subsets following the model of the fourth example above. This argument may appear untenable if the FPI technique is conceptualized in terms of its semiclassical approximation, in which paths through barriers (regions of very large potential) are simply ignored. (This approximation method is *distinct* from the “finite representative sampling” technique mentioned above.) In an exact implementation of the FPI technique, however, even paths through barriers must be summed over. This approximation technique is valid for the FPI since nearby neighboring paths through regions of large potential will tend to cancel each other out in the summation. (See section 2-3 of [24] for an excellent presentation of this issue.) The paths that are typically omitted in the semiclassical approximation will in general vastly outnumber the paths that are typically included. Compare, for example, Figure 2 E to Figure 2 F. The former represents Feynman paths for a hypothetical experiment generated via the semiclassical approximation, with two paths 120 degrees separated. The latter represents the same experiment in which twelve additional paths through a barrier are also included. Since these twelve paths traverse a region of high potential, then the actions of neighboring paths will tend to vary greatly, the effect being that they will tend to be “randomly” distributed around the unit circle and will therefore cancel each other out when summed together via the FPI formalism. Note that the diagram of Figure 2 E, when viewed through the MMWI formalism, implies that $\bar{\chi}_{s,d_n}^1 = 2$. However, when the additional paths of Figure 2 F are included, it is seen that $\bar{\chi}_{s,d_n}^1 = 1$. The end result is that the Feynman wavefunction Ψ (depicted in Figure 2 E) is recovered

using the MMWI formalism as the quantity $\bar{\chi}^1$ (depicted in Figure 2 F) provided that paths through barriers are included for the determination of $\bar{\chi}^1$. A more general statement would be that it is always possible to categorize paths into subsets following the model of the fourth example above. Thus, given any experimental setup and an ensemble E of manifolds W , there will always exist a manifold $W \in E$ in which the minimum value of χ_{s,d_n}^1 is achieved, $\bar{\chi}_{s,d_n}^1 = \chi_{s,d_n}^1$, and in which $(\chi_{s,d_n}^1)^2 \sim |\psi_{s,d_n}|^2$. In contrast to the “finite representative sampling” technique, it is argued that this semiclassical approximation technique should typically *not* be utilized within the MMWI, as it would frequently obviate the argument presented in this paragraph – as illustrated by Figure 2 E.

Finally, consider the second stage of the diagram of Figure 1 E. By experimental design, the particle traverses the source s as well as one and only one of the detector regions d_n . Given any two regions a and b , and given that a and b correspond to two different spacetime locations of the same 4-geon, it is assumed that there exist *two* index $k = 1$ paths from a to b that can be uniquely associated with the 4-geon. (This assumption is discussed further in Sec. VIC.) Given $\bar{\chi}_{s,d_n}^1$ paths of index $k = 1$ from s to d_n , there are exactly $\frac{1}{2}(\bar{\chi}_{s,d_n}^1) * (\bar{\chi}_{s,d_n}^1 - 1) \sim (\bar{\chi}_{s,d_n}^1)^2$ unique ways to choose two of them. Assume that in the second phase of the trajectory diagram (Figure 1, E), the observation of the particle at detector element d_n is concomitant with identification of these two paths, and that there is a separate branch on the trajectory diagram associated with each possible pairwise combination. Thus, the number of branches associated with the n^{th} detector element is proportional to $(\bar{\chi}_{s,d_n}^1)^2$. By the EPP, the predicted probability P_{s,d_n}^{MMWI} of detection at d_n is proportional to this number of branches, so that $P_{s,d_n}^{MMWI} \sim (\bar{\chi}_{s,d_n}^1)^2$. The FPI, of course, predicts that $P_{s,d_n}^{FPI} \sim |\psi_{s,d_n}|^2$. Since $(\bar{\chi}_{s,d_n}^1)^2 = |\psi_{s,d_n}|^2$, it follows that $P_{s,d_n}^{MMWI} \sim P_{s,d_n}^{FPI}$; that is, the predictions of the MMWI are in agreement with the predictions of the FPI in the analysis of a single particle. Since the FPI makes exactly the same predictions as *any* of the other formulations of quantum mechanics [28] – e.g., matrix mechanics, wave mechanics, second quantization, the de Broglie-Bohm pilot wave formulation, etc. – it may be concluded that the MMWI is equivalent to any of these formulations, and thus to quantum mechanics in general.

To address the difference between bose and fermi statistics from the framework of the MMWI, it will be necessary to analyze a problem involving the emission of two (or more) particles which interact between emission and detection. Since this scenario is not represented in the analysis of Sec. V, it must be admitted that the difference between bosons and fermions is yet to be addressed, and so it cannot be said that the *whole* of quantum statistics has in fact been demonstrated. Preliminary work, however, indicates that if fermions, but not bosons, are postulated to make an “exchange” of a portion of their 3-volumes a and b during their interaction, then bose statistics and fermi statistics may be demonstrated in the resulting theory. Indeed, the exchanged portion in this construction would itself be a boson, fulfilling the role of force-carrier between fermions. This will be the subject of a future paper.

VI. FURTHER MATHEMATICAL CONSIDERATIONS

There are obviously many mathematical issues that must be addressed for the the MMWI to be developed further. Several of these will be mentioned, albeit only briefly, in this section.

A. Observer state space

The first issue is to provide a more detailed description of the mathematical state of the observer. In principle, the essential ontology of the MWI allows the role of the observer to be assigned to *any* physical object: a computer, a human being, or even a particle. To define a state space for the observer, however, it would perhaps be helpful to choose a more restricted definition of the observer according to some convention that makes the further mathematical development of the scheme more tractable. It is worthwhile to note, therefore, that if spacetime is assumed analytic, then the method of analytic continuation provides an exact solution for the spacetime metric g_{ij} *throughout an entire spacetime region*, given only the value of the metric and its power series (all of its derivatives, including partials) *at a single point* within the region. For this solution to be unique, however, the topology of the region must be prespecified. For the sake of convention, therefore, if the observer ψ^O is assumed to correspond to, say, a simply-connected region of spacetime, then the observer state may be given full representation by specifying simply the metric g_{ij} and its power series at *one point* p inside ψ^O . Although there are only ten components to g_{ij} , there are of course an infinite number of components involved in providing its power series. It follows that observer state space is infinite-dimensional – as might be expected, given the infinite dimensionality of standard quantum mechanical Hilbert space. It is intriguing to speculate that these ten components might in some way correspond to the ten dimensions that play a central role in many versions of string theory – although this is nothing more than a speculation at this point.

Using these conventions, the y-axis of the world-splitting diagrams of Figure 1 corresponds to the location of the observer in this infinite-dimensional state space. A choice of the “forward time direction” t (which by convention is a required part of the definition of the observer) defines a vector at p . The evolution of the observer in time (along the x-axis of the Figure 1 diagrams) is calculated by parallel transport of this vector by the parallel-transport law, i.e. according to the geodesic equations [29]:

$$\frac{d^2 x^k}{dt^2} + \Gamma_{ij}^k \frac{dx^i}{dt} \frac{dx^j}{dt} = 0 \quad (1)$$

In other words, the trajectories in the world-splitting diagram correspond to *geodesics* through spacetime. Since there are a multiplicity of manifolds W in the ensemble E (with distinct topologies), there will be a multiplicity of geodesics. If spacetime is assumed to be governed by Einstein’s equation, then the evolution of the observer is likewise governed by Einstein’s equation.

B. Lattice approximation of W

A second issue is to provide an explicit mapping from each of the $W \in E$ onto the “approximation” manifold M . A lattice approximation technique may be useful for this purpose, whereby a lattice of points is distributed throughout W with p (as defined above) at the origin. This lattice may then be used as the approximation manifold M itself. The manifolds W are assumed to be multiply connected *at the small scale* in a fashion reminiscent of spacetime models of “quantum foam” as proposed by Wheeler [30], Hawking [31], and others. That is, W is assumed to be violently warped and twisted on the small scale but not on the large scale. Therefore, the flat manifold M can be imagined to be constructed by “ironing out” the small-scale curvatures of a given W . In addition, the 3-dimensional regions a and b in W are assumed to be small enough to be approximated by points a' and b' in M . This is necessary to apply the FPI technique to M .

If the lattice is sufficiently coarse-grained – that is, constructed such that the distance between adjacent points is sufficiently large (larger than, say, the size of a typical “quantum foam” wormhole in W) – then the lattice may be constructed so that it is simply connected, 4-dimensional, and flat (approximately equal spacing between adjacent points). The set of all paths φ connecting two points a' and b' in M may then be constructed as the set of all sequences of points on the lattice connecting a' to b' . Likewise, a set of “representative” paths λ connecting a and b in W (which is a *subset* of *all* paths λ) may be constructed in a similar manner, i.e. by considering all paths generated as a sequence of points in the lattice, and considering only those λ that are homotopic to those paths. As the lattice becomes more fine grained, the approximation manifold M changes (first) from flat to curved, and (next) from simply to multiply connected; in the limit as the spacing between adjacent points approaches zero, the lattice resembles W more and more. Indeed, it may also be noted that the underlying structure of the world-splitting diagram may be envisioned to transition in the fine-grained limit from *discrete* to *continuous*.

Whether such a transition is expected to exist, however, is a subject for future work. Recall that the considerations of Sec. V make use of a *single* manifold M for the approximation of *every* $W \in E$. (The notion that the mapping from every $W \in E$ onto a lattice will yield the *same* energy distribution throughout the points on the lattice – at least in the proximity of the experimental apparatus – is essentially equivalent to the notion that every $W \in E$ contains the *same* experimental setup, as discussed earlier.) The use of the non-relativistic FPI technique to approximate the calculations of the MMWI, therefore, will require a lattice that is sufficiently coarse-grained for M to remain flat and simply connected.

A 4-volume of spacetime in W may be defined surrounding each point in the lattice, the volumes being disjoint and the union of all such volumes covering the entire spacetime W . Thus, the lagrangian at a point on the lattice may be calculated from W as being in some sense equivalent to the lagrange density of the associated 4-volume in W . The action of path φ in M is then calculated as the sum of the lagrangian of each

point along the sequence of points that define the path. In W , a suitable method to calculate the action of path λ may be, not as a line integral, but rather as a 4-volume integral of the lagrange density, with the volume defined as the union of 4-volumes of spacetime corresponding to the sequence of points in the lattice corresponding to the path. In this way, the action of path λ in W calculated as a volume integral of the lagrange density is necessarily equal to the action of path φ calculated as a line integral of the lagrangian in M . This is an important conclusion, since the equivalence of the action of λ with the action of its corresponding path φ is one of the necessary assumptions for the considerations of the previous section.

C. Topological properties of a 4-geon as a basis for the “cyclic ordering” of the amplitudes of paths

Much work is needed in the further specification of the mathematical character of a 4-geon. As discussed above, a 4-geon at a particular point in time is a (small) 3-dimensional volume of spacetime a which is to be considered a hypersurface or “time-slice” through a 4-dimensional region that defines the entire existence through time of the 4-geon. Within a given W , the trajectory of a 4-geon from a to b may be envisioned as a long thin piece of string, with a being a slice through the string. However, it will be proposed that the natural global topology of a 4-geon is that of a torus, with a and b being two hypersurfaces within this torus. Thus, the “God’s eye view” of a 4-geon is not a piece of string, but rather a *loop* of string through 4-dimensional spacetime. From this perspective, the two unique $k = 1$ paths connecting a and b may be envisioned to correspond to the two ways around this loop of string.

It may further be postulated that the topology of any hyperslice through this torus is preserved, so that there is a one-to-one correspondence between the CTCs in a and in b . In this manner, the notion that the placement of paths into subsets is preserved between two different regions of the same 4-geon may be seen to be more feasible. Indeed, this detail for the model of a 4-geon may be further utilized to justify the assumption of the cyclic ordering of the amplitudes of the k paths within a subset. Consider the diagram of Figure 3 A, which depicts two hypersurfaces a and b through a 4-geon. Following Hadley, these hypersurfaces are assumed to be multiply connected and to contain many CTCs, one of which is depicted as a green loop in each hypersurface. It is envisioned that paths λ from a to b may be placed into subsets by virtue of their mutual coassociation with a single CTC. Figure 3 A depicts coassociation of five λ s by a CTC in a ; of note, it is assumed that these *same* five paths are mutually coassociated into a $k = 5$ subset via a corresponding CTC at b . Consider next a 1-form w over W , defined so that the integral of w yields an action S . A central assumption will be that w can be expressed as the pull-back of the standard 1-form on the unit circle. This implies that the integral of w along any closed loop C will be integer-valued. In other words, the integral of w around any given CTC inside a time-slice of a 4-geon takes

the form:

$$\oint_{CTC} w = \pi \kappa n \quad (2)$$

with κ being some constant, and n being an integer, which is assumed to be unity $n = 1$ for the CTCs inside a geon hypersurface. Finally, the k points of connection between the CTC and the k paths λ are assumed to divide the CTC into k segments of equal action $\pi \kappa / k$, as depicted in Figure 3 A.

An interesting feature of this construction is that κ suggests itself as a natural “measuring stick.” In other words, the “minimum distance around a CTC” could in fact serve as a natural “clock and ruler” by which distances are *defined*. Since κ by definition comes in units of action S , it is natural to use κ as a *definition* of Planck’s constant, $\hbar$ (which also has the units of an action S): $\hbar = \kappa$.

The action corresponding to a loop that stretches between a and b , such as the loop Λ depicted in Figure 3 A, is assumed to take the form:

$$\oint_{\Lambda} w = 2\pi \kappa m \quad (3)$$

with m being any integer. Keeping in mind that paths are directional (so that the integral of w along a path in one direction is the negative of the integral in the reverse direction), the above assumptions serve to derive the cyclic ordering of the amplitudes of paths, in the following manner:

Theorem 1. *Cyclic ordering of the amplitudes of paths. Any two adjacent paths λ^i and λ^{i+1} that are mutually coassociated within a subset of critical index k will be $2\pi\hbar/k$ out of phase.*

Proof.

$$\oint_{\Lambda} w = 2\pi \hbar m$$

$$\int_{x_1}^{x_2} w + \int_{x_2}^{x_3} w + \int_{x_3}^{x_4} w + \int_{x_4}^{x_1} w = 2\pi \hbar m$$

$$\pi \hbar / k + S^i + \pi \hbar / k - S^{i+1} = 2\pi \hbar m$$

$$S^i - S^{i+1} = 2\pi \hbar m - \frac{2\pi \hbar}{k}$$

□

The significance of this theorem is that a seemingly ad-hoc assumption, the cyclic ordering of the amplitudes of paths, has been replaced with a field-theoretic assumption regarding the pull-back of the 1-form w . This is an example of the “reverse engineering” approach that has been utilized for the development of the MMWI. Work is in progress to investigate whether a connection may exist between this field-theoretic statement and Einstein’s equation.

D. Minimization laws and the micro-structure of the world-splitting diagram

The micro-structure of the world-splitting diagram that is postulated in Figure 1 E plays a central role in the construction of the MMWI. A justification for this micro-structure from deeper principles is therefore necessary to place the MMWI on a firm footing.

The observation of a physical parameter is defined as a branching event in which different branches are associated with different values of the physical parameter. One may ask whether any given branching event should occur “all at once,” as on the right-hand side of Figure 1 E, or rather over some extended time interval, as on the left-hand side of Figure 1 E. It should be evident with some contemplation that this should depend upon whether the physical parameter in question is a property of an extended region of spacetime, or is a property of a point in spacetime. An example of the latter would be the value of the metric, which is formally a local property. Examples of the former would be certain topological properties, such as the genus of a region, which is formally a *global* property that is not “encoded” in any one point.

For the observer to measure a property of an object at some instant of time, relativity demands that the object must be within the past light cone of the observer. This notion is depicted in Figure 3 B, where a corresponds to one hypersurface (a time-slice) through a 4-geon. The y-axis of the diagram corresponds to the time axis t in the reference frame of the observer, and the x-axis of the diagram corresponds to the remaining spatial dimensions, with the observer at location $x = x_0$. Suppose that the parameter in question is the number η of CTCs inside the region a ; each CTC is represented as a black dot. At time t_1 , the region a is not in the past light cone of the observer, and so no property of a has been communicated to the observer. By time t_3 , a small portion of a lay within the past light cone; therefore, the observer can have counted up the CTC’s inside this small portion. The process of the observation of η may then be thought of as occurring over the time interval from t_2 to t_4 . At any given point in time between t_2 and t_4 , the observer “knows” that η is *at least* equal to the total number of CTCs that have so far fallen within the past light cone. It is as if the observer asks, at each instant in time: is η equal to the number of CTCs that have been counted so far, or are there more? By the EPP, each iteration is represented by a bifurcation, with each of the two branches being realized with 50 percent probability. The resulting world-splitting diagram corresponding to the observation of η will thus resemble that on the left hand side of Figure 1 E corresponding to the observation of χ^1 . At any given instant of time, if the entire region a has fallen within the past light cone of the observer, then the observer may conclude that η equals the number of CTCs that have been counted up so far. If there is still any portion of a that is not in the past light cone, then η may be larger.

The result of the proposed micro-structure to the world-splitting diagram for the observation of η , when the EPP is applied, is that η – the number of CTCs within a – is minimized. Consider, once again, the discussion of the preceding

section regarding the minimization of χ^1 . Given a set number of paths λ_{s,d_n} , it may be remarked that the minimization of η *implies* the minimization of χ^1 . This is because a smaller number of CTCs available to a constant number of paths λ implies a greater number of paths to have high index k , and – likewise – a lesser number of paths to have small index such as $k = 1$. Therefore, the assumption that the micro-structure of the world-splitting diagram on the left of Figure 1 E takes the form that it does, such that χ^1 is minimized, may be *replaced* by the assumption that, rather, η is minimized. This assumption, in turn, is justified via the observation that η is, by definition, a global topological property.

These considerations are, admittedly, heuristic, and far from rigorous. Nevertheless, this may be considered an illustration of how a *minimization law* may arise within the context of the MMWI. One of the more beautiful aspects of the FPI is the ease with which the principle of least action may be seen to arise in the classical limit of the quantum laws [24]. Since the quantum laws are shown in this paper to arise from the Born constraints, the (classical) principle of least action may be seen to arise more fundamentally as a result of the fact that η is a topological property. That is: η is a topological property $\rightarrow$ principle of least $\eta \rightarrow$ principle of least $\chi^1 \rightarrow$ principle of least action. This is, of course, merely a heuristic argument. Nevertheless, it would be interesting to explore whether other classical minimization laws may be generated in a similar fashion. The beauty of this argument is that a minimization principle may be understood to be rooted fundamentally in a symmetry principle: the EPP.

VII. DISCUSSION

The purpose of this paper has been to demonstrate, by example, that it is possible to devise a variation of the MWI that is based on the EPP, that makes predictions in agreement with quantum mechanics, and that preserves the full ontology of the MWI. In addition, the proposed scheme is founded upon a field-theoretic description of matter and is thus proposed as a means for the derivation of quantum statistics from an underlying field theory. Perhaps the biggest ontological “pill” to be swallowed is that the underlying field theory is assumed, via the 4-geon particle model and the heavy use made of CTCs, to be non-causal on the small scale. This view, however, is predicted by many quantum foam models, and may be viewed as a small price to pay for a derivation of quantum statistics from the EPP.

Indeed, many aspects of the present scheme have been guided by ontological considerations. In an attempt to make some of these ontological issues more apparent, the MMWI has been presented in this paper via a series of stepwise modifications to the SMWI, so that the ontological impact of each modification may be considered at each step. First, the aesthetic advantages of the probability criterion are incorporated by replacing the Born rule with the EPP. In this case, the MMWI, perhaps more than the SMWI, respects the original ontology of Everett’s programme, according to which each world (or equivalently, each branch) is attributed an equal de-

gree of “reality.” Alternately, the EPP may be justified via an argument by symmetry, independent of the ontological “reality” of parallel worlds. Next, the ability to make empirically correct predictions is restored by replacing the standard enumeration postulate (SEP) with an alternate enumeration postulate (AEP) that obeys the Born identity, as illustrated in Figure 1 B. Although the SEP – that is, the one-to-one correspondence between the N_O states of the observer and the N_S states of the system under observation – is made, at least implicitly, in Everett’s original paper, it is argued that the SEP is *not* an *essential* aspect of Everett’s original programme. Indeed, one may imagine a version of the “many minds” variation [16] of the MWI according to which each branch in Figure 1 B represents one “mind.” Although this interpretation has the advantage of being consistent with the EPP as well as with quantum statistics, the difficulty is that there is no way to distinguish among the J_n minds that make an identical observation and thus no *physical* explanation upon which to understand this overall scheme. To remedy this, the proposed scheme is based upon the assertion that the world-splitting diagram, *by definition*, is a reflection of the physical state of the observer, and that wherever branching occurs, each branch must represent *physically distinct observer states*. Therefore, the MMWI makes sure to distinguish different branches of the world-splitting diagram by virtue of a physical distinction between different observer-states – once again, in a manner that respects the ontology of Everett’s original programme.

The purpose of Figure 1 C is to emphasize the fact that, at this point in the conceptual development of the current scheme, there are likely *many* variations in the scheme outlined in Figure 1 B that could perhaps meet the ontological criteria discussed so far. These criteria would permit, for example, a scheme that incorporates *fusion* as well as splitting in these diagrams (although fusion has not been explicitly utilized in the present scheme). Indeed, the model of Figure 1 C may be viewed as a sort of “empty canvass” upon which to build a more detailed model that respects the aforementioned ontological criteria. It should therefore be emphasized that the further details of the proposed scheme as outlined in Figure 1 E – including the 4-geon particle model, the reliance on general relativity, and the Born constraints – are offered as an “existence proof” for the notion that it is at least *possible* to take this “empty canvass” and flesh out the details into a fully formed programme. Alternate strategies may exist for the same purpose.

In the proposed scheme, the next step for filling in the details of this “empty canvass” is to provide a mathematical description of the observer. Several arguments have been presented that motivate a general relativistic description of matter in terms of 4-geons for this purpose. The first is the existence of a “multiplicity” of solutions to the time-dependent evolution of a general relativistic system that admits CTCs. One of the arguments that is often put forward for the incompatibility of GR and QM is that QM is fundamentally a theory of probability, whereas GR is a deterministic theory that has no inherent concept of probability. This argument is challenged, however, by the appearance of multiplicity within GR, provided that one accepts the EPP as a natural probability inter-

pretation of multiplicity. Next, Sec. VID provides a heuristic argument that general relativity, coupled with the EPP, provides a natural foundation for the derivation of minimization laws. Finally, it may be argued that the success of general relativity in its own right makes it simply a *worthy* foundation upon which to attempt to build the MMWI. It should be noted, however, that there is no *a priori* reason that some other field theory, such as the Evans unified field theory [23] or perhaps some version of string theory, may serve as an alternative to GR for this purpose.

A final ontological feature of Everett’s programme should be noted. The MMWI makes use of a relationship between the observer and system under observation that is consistent with, albeit perhaps only implied in, Everett’s original paper: that is, the notion that the state of the system ψ^S , and indeed of the entire external world, is *determined* by the state of the observer ψ^O . The exploitation of this particular feature of the MWI via a mathematical approach similar in spirit to the holographic principle underscores the hidden riches of the Everett programme that are perhaps yet to be fully appreciated. In the MMWI, this feature takes expression via the use of ψ^O as a boundary condition for the 4-manifolds W . The fact that W is not determined *uniquely* by ψ^O is reflected in the notion of an ensemble E of manifolds that meet this boundary condition. Indeed, this notion of the ensemble of manifolds may be regarded as giving rise to the notion of *superposition*. A discussion of the relationship between the MMWI and other ensemble formulations of quantum mechanics, such as Einstein’s “statistical formulation,” will be the subject of a separate paper.

Great care has also been taken in the construction of the MMWI to respect established criteria of locality and non-locality. The fact that the MMWI gives rise to quantum statistics should ensure that Bell- (that is, EPR-) nonlocality is a general feature of the MMWI. An analysis of a typical Aspect experiment using the formalism of the MMWI will be the subject of a separate paper. A more difficult matter has been to ensure that Einstein-locality is also respected by the MMWI. The general argument that the MMWI respects Einstein locality is, simply, that each of the 4-manifolds W are assumed to follow the restrictions of general relativity. It should be noted, however, that the construction of the world-splitting diagram does *not* require, at any point, non-local transfer of information, as was pointed out in Sec. V. Consider the fact that the identity of different branches is associated with different values of some physical parameter – such as χ . Great care has been taken to ensure that the values of such physical parameters *are within the past light cone of the observer at the time that the branch occurs*. The dictates of Einstein-locality and Bell-nonlocality as they apply to the MMWI will also be the subject of a separate paper.

An alternate way to conceptualize the MMWI, aside from thinking of it as a modified version of the SMWI, is to view it from first principles and see how quantum statistics is derived. This derivation is envisioned to proceed as follows: general field-theoretic constraints $\rightarrow$ Born constraints $\rightarrow$ Born identity $\rightarrow$ quantum statistics. The last step – quantum statistics from the Born identity – is a straightforward application of the

EPP. Indeed, it should be noted that the goal of deriving quantum statistics from relativity is worthwhile only if one accepts the EPP as being sufficiently natural as to require no further justification. In other words, it would be pointless to attempt a derivation of quantum statistics from an underlying field theory in the context of the SMWI, since the SMWI assumes the Born rule, and as such, it assumes quantum statistics already. The MMWI, on the other hand, is envisioned as a programme that does not assume quantum statistics, but rather derives it from deeper principles.

This paper is concerned primarily with the second step: the approximation of the Born constraints by the Born identity. The Born constraints may be regarded as a loose set of criteria falling into two categories: first, a model of 4-geons; and second, a set of relations applicable to the paths λ connecting any two regions of spacetime a and b within a 4-manifold M . Using this framework for the analysis of a particle emitted from a source s , it is shown, first, that the MMWI predicts the probability of observation by the n^{th} of N detector elements to be proportional to $(\bar{\chi}_{s,d_n}^1)^2$, and second, that this quantity is proportional to the relative probability $|\Psi_{s,d_n}|^2$ as calculated by the Feynman path integral. Since the FPI is equivalent to any of the other accepted formulations of quantum mechanics [28], it may be concluded that the MMWI makes the same predictions as quantum mechanics in general.

Although the above-mentioned second category of Born constraint criteria is outlined in loose form in this paper, it is envisioned that they could be reformulated as more compact mathematical statements – ultimately derivable from some underlying field theory, such as general relativity. There are many mathematical languages for the expression of general relativity (such as differential geometry, differential forms, or geometric algebra), and it is unclear to the author which might be the most appropriate in a search for the derivation of the Born constraints from Einstein’s equation. Sec. VI is intended to give a flavor for how this work might proceed. Given a model of 4-geons and a 1-form w whose integral is interpreted as an action, the assumption that w can be expressed as the pull-back of the standard 1-form on the unit circle is shown to provide the basis for a significant portion of the Born constraints. The crucial question remains, then, whether some sort of relationship might be shown to exist between this assumption and Einstein’s equation. If this can be done, then it may be possible to *derive* much of the formalism of the MMWI directly from general relativity itself.

The lattice approximation method discussed in Sec. VIB is provisional only. A central issue to be addressed is whether the probabilities generated by the MMWI formalism remain stable as $M \rightarrow W$, that is, as the lattice approximation becomes (infinitely) more fine-grained. Indeed, this issue may be the MMWI-equivalent of the renormalization difficulties inherent to most if not all of the standard quantum mechanical formalisms. If the proposed scheme proves to be overall feasible, then this issue could, perhaps, shed new light on how to address the issue of renormalization in general. A related issue is whether the method for the calculation of the action of the paths λ and φ , proposed in Sec. VIB, can be made mathematically rigorous. Furthermore, this must be done in a

manner consistent with the definition of w and *its* interpretation as an action, as outlined in Sec. VIC.

Another important remaining question is to specify explicitly the operator for the time-dependent evolution of the observer. As outlined in Sec. VIA, the observer exists as a point in an infinite-dimensional state space, which is interpreted as the power series of the metric g_{ij} at a physical point in spacetime. This operator must describe the parallel-transport of the observer through a 4-dimensional manifold, which is presumed to be governed by Einstein’s equation. Since Einstein’s equation is nonlinear, it may be thought that this operator must also be nonlinear; if so, then nonlinearity is a feature that the MMWI shares in common with the scheme by Weissman [7]. Further work, though, is needed to address this issue. A related question is whether the state space of the observer is a Hilbert space. By analogy to standard formulations of quantum mechanics, it might be expected that the answer to this question is “yes.” However, quantum mechanics is proposed in this paper to be valid as an *approximation* to a (possibly nonlinear) underlying field theory, so it remains to be seen whether the Hilbert space structure of state space should carry over.

The proposed scheme, in its present form, is insufficiently developed to propose an experimental test between it and the standard formalism. If new predictions are to be made within the context of the proposed scheme, then it is likely to be within the realm of particle physics, through a more detailed description of the inner structure of particular 4-geons – via, for example, a further investigation of the difference between the most important particle classification, fermions and bosons. As further subclassifications are investigated, perhaps experimental tests will suggest themselves. In the absence of such tests, then, one may ask whether the interpretational advantages of the proposed scheme are worth the price of a more complicated mathematical machinery. From one perspective, this is a matter of “taste.” However, it may be argued that the mathematical machinery of the proposed formalism is complex precisely because it is ultimately founded upon a field-theoretic description. The present formalism, therefore, is no more complex than any of the existing quantum gravity formalisms. In the proposed scheme, quantum statistics is seen to be intimately connected, not only to the internal structure of fundamental particles, but also to more global spacetime constraints. Indeed, it would perhaps be a surprise (or at least a disappointment) if a theory of quantum gravity were to be devised that did *not* reveal such a connection.

Acknowledgments

Blair Smith, Robin Hanson, Michael B. Weissman, Patrick Van Esch, Ed Huff, Mark Hadley, Carl Brannen, Steve Harris, Scott C. Johnson, Thomas D. Gutierrez, and David Charles Hobby are warmly thanked for their insights and discussions.

APPENDIX A: RELATIONSHIP BETWEEN THE PROBABILITY CRITERION AND THE EPP

Assume the definitions of Sec. II. It is further assumed that m_n can be expressed as a function of N . (Compare this to Everett's assumption [1], discussed in [6], that m_n can be expressed as a function of a_n .) The goal is a solution for m_n that satisfies the probability criterion.

The number of ways to make P_n observations of the n^{th} outcome is calculated to be $\frac{(N-1)^{(M-P_n)} M!}{P_n! (M-P_n)!}$, as follows. There is only one way to get the n^{th} result on P_n out of P_n trials. Next, the number of ways to get anything other than the n^{th} result on $M - P_n$ out of $M - P_n$ trials is $(N-1)^{M-P_n}$. Next, the number of ways to mix an ordered sequence with P_n elements and an ordered sequence of $M - P_n$ elements, i.e. the number of ways of distributing P_n elements over M elements is M choose P_n , i.e. $\frac{M!}{P_n! (M-P_n)!}$. Multiplying these expressions yields: $\frac{(N-1)^{(M-P_n)} M!}{P_n! (M-P_n)!}$. Dividing this expression by the total number of worlds N^M yields the proportion $f(p_n) \in [0, 1]$ of such worlds: $f(p_n) = \frac{(N-1)^{(M-P_n)} M!}{P_n! (M-P_n)! N^M}$. The integrated proportion $F(m_n, \delta) \in [0, 1]$ of worlds in which the observed frequency p_n is close to the predicted frequency m_n , i.e. falls anywhere within the closed interval $m_n \pm \delta$, i.e. falls within $[m_n - \delta, m_n + \delta]$, is calculated as the sum: $F(m_n, \delta) = \sum_{p_n=m_n-\delta}^{p_n=m_n+\delta} f(p_n)$. The probability criterion states that for any δ , $\lim_{M \rightarrow \infty} F(m_n, \delta) = 1$. It is readily seen that the EPP, $m_n = 1/N$, is a solution to this equation. Computerized numeric calculations confirms this solution. Therefore, it can be concluded that the EPP is a solution to the probability criterion.

APPENDIX B: REVIEW OF THE FPI

This section presents a compact overview of the basic steps involved in the application of the Feynman path integral technique to the solution of the following very general problem: given a particle emitted by a source $s(x_1, x_2, x_3, t)$,

calculate the probability of its detection by detector element $d_n(x_1, x_2, x_3, t)$. See Ref. [24] for details.

Step 1: Draw every path φ from s to d_n . In principle, this technique assumes an infinite number of paths. However, a common method of approximation is to rely instead on the “ M most representative” paths, where M is a (suitably large but finite) integer, so paths are labelled with the integer $m \in [1, M]$.

Step 2: For each path, define the lagrangian $L = KE - PE$ at each point on the path (where KE is the kinetic energy and PE is the potential energy), and use this to calculate the classical action S associated with the path:

$$S_{s,d_n}^m = \int_{\varphi_{s,d_n}^m} L \quad (B1)$$

Step 3: Use the action to calculate the amplitude ϕ :

$$\phi_{s,d_n}^m = A e^{-i S_{s,d_n}^m / \hbar} \quad (B2)$$

where A is a normalization constant. These amplitudes may be plotted on the complex plane as in Figure 1 D or Figure 2.

Step 4: Calculate the wavefunction Ψ_{s,d_n} that the particle will be detected at d_n as the sum of all the amplitudes:

$$\Psi_{s,d_n} = \sum_{m=1}^M \phi_{s,d_n}^m \quad (B3)$$

Step 5: Calculate the relative probability P_{s,d_n} that the particle will be detected at d_n :

$$P_{s,d_n} = |\Psi_{s,d_n}|^2 \quad (B4)$$

Step 6: For each of the N detector elements d_n , repeat steps 1 through 5.

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